

Quiz 4

Name: _____ Roll No. _____

1. Is it possible to have **only** 2 pure strategy Nash Equilibrium in the following game?

Explain next to the table.

2 marks

		Player 2	
		A	B
Player 1	A	a, a	b, b
	B	d, d	c, c

Answer. For sure. In fact, such games are quite common, and are a class of coordination games. Can explain with an example.

		Player 2	
		A	B
Player 1	A	3, 3	0, 0
	B	1, 1	4, 4

Other examples are okay too, or if you just explain it using the parameters it is okay.

2. Consider a modified Prisoner's Dilemma. Suppose Player 1 went first, then player 2 observed Player 1's action and went second, and then player 1 went again (this time without observing player 2's action). Both have actions C and D as in the classic game. Write the strategies for each player. To make it a little easier, consider player 1's actions the second time as C' and D'

2 marks

Player 2 is easy but player 1 is a little tricky. Since player 1 knows whether she has chosen C or D in the beginning, but cannot observe player 2's action, she has 3 decision points for a strategy. It is like the centipede game, where P1 strategises for In even when he plays Out.

Player 1: $CC'C', CC'D', CD'C', CD'D', DC'C', DC'D', DD'C', DD'D'$

Player 2: CC, CD, DC, DD

Order may change based.

3. Suppose we consider the goalkeeper vs kicker game (both world class). Realistically, the GK can go center (C) right up (RU), right down (RD), left up (LU) and left down (LD). But the kicker can kick anywhere on the rectangle that is the goal. (Suppose the ball is just a point). The length of the goal is $l > 0$ and the height is $h > 0$. Represent the actions for each player in a mathematically accurate manner. **3 marks**

GK: $a_{GK} = \{C, RU, RD, LU, LD\}$

Kicker: $a_{Kicker} = [0, l] \times [0, h] \subset \mathbb{R}^2$

If you don't write $\subset \mathbb{R}^2$ only 1 mark is deducted

XXXXXXXX

A less fun but equally accurate representation could be

(x, y) where (or $|$) $0 \leq x \leq l$ and $0 \leq y \leq h$

XXXXXXXX

Finally, (x, y) where (or $|$) $x \in [0, l]$ and $y \in [0, h]$

also works

4. The primary concern with vocational training via Industrial Training Institutes (ITIs) is **1 mark**

(A) Quantity (B) Quality (C) Slow increase in number (D) NOTA

Answer. (B) Quality. Easy

5. Oil prices are included in

(A) Core inflation (B) Non-core inflation (C) Headline inflation

(D) More than one of these. **1 mark**

Answer. (D). Both (B) and (C) so answer is more than one.

6. Suppose the action set of 2 players is whoever chooses the larger natural number wins.

How many Nash equilibrium does this game have?

(A) 1 (B) 3 (C) ∞ (D) 0

1 mark

Answer. (D) This is a case of no Nash Equilibrium (non-compact action space)

Clarifications (Does not apply if correct answer is definite from the question).
